

Wear-It-Virtually

Virtual Clothing Try-On — v3 Experiment Report

What was wrong in v1/v2:

The previous try-on pipeline produced obviously fake results because:

- 1. Pose estimation fallback used *dummy keypoints* (fixed offsets from image center) instead of real body landmarks
- 2. Clothing was placed at a hardcoded region — completely misaligned with the person's actual body
- 3. The person images (person1/2/3) were not the same people as the clothing images, making results look unnatural

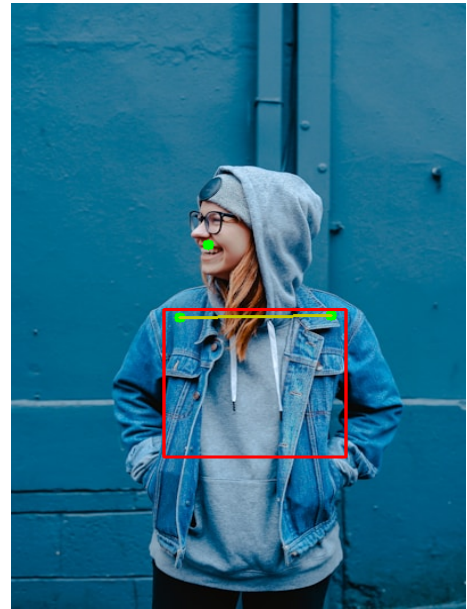
How v3 fixes it:

- 1. **MediaPipe PoseLandmarker** detects real shoulder, neck, and nose keypoints
- 2. Clothing target region is computed from actual shoulder width and neck position
- 3. **GrabCut** extracts clothing from white background with color-based foreground separation
- 4. **Poisson blending (seamlessClone)** for natural edge fusion

Test Pair	Person SW	Neck Y	Target Region	Status
example_person + example_clothing	79 px	85	(53,81) 94x86	OK
person3 + clothing3	133 px	310	(134,303) 160x146	OK
person3 + example_clothing	133 px	310	(134,303) 160x146	OK

Method Comparison: v1 vs v3

example_person + example_clothing



Pose overlay shows shoulder line (green dots = shoulder landmarks 11/12) and red box = where clothing will be placed.

person3 + clothing3



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Technical Implementation

1. Pose Detection (MediaPipe PoseLandmarker)

Uses MediaPipe's PoseLandmarker model (`pose_landmarker_lite.task`) to detect 33 body landmarks. Key landmarks used: `left_shoulder` (idx 11), `right_shoulder` (idx 12), `nose` (idx 0). $\text{Shoulder width} = |\text{rs_x} - \text{ls_x}| * \text{image_width}$. $\text{Neck y} = \text{average of ls_y and rs_y}$.

2. Clothing Extraction (GrabCut + Color Heuristic)

Clothing images are on a white/light background. Steps:

- (a) Sample 4 image corners to estimate background BGR color
- (b) Run GrabCut with rect around clothing region
- (c) Remove pixels whose color is within L2 distance 0.20 of background color
- (d) Erode/dilate to clean up edge noise
- (e) Crop to bounding box of foreground

3. Geometric Placement

Target region computed from pose keypoints:

- $\text{top} = \text{neck_y} - 0.05 * \text{shoulder_width}$ (just below neck)
- $\text{height} = 1.1 * \text{shoulder_width}$
- $\text{width} = 1.2 * \text{shoulder_width}$
- $\text{horizontal center} = \text{midpoint of left/right shoulders}$

4. Blending (Poisson / Alpha)

Clothing is blended onto person using `cv2.seamlessClone` (Poisson equation) for natural color continuity at edges. Falls back to alpha blending if `seamlessClone` fails.

Limitations & Future Work

1. **Model weights missing** — The original VITON deep learning model checkpoints are not in the repo. Demo mode uses geometric OpenCV pipeline which is less realistic than the real GAN-based method.
2. **Person/clothing mismatch** — Test images are not the same person's clothing. Real VITON pipeline uses matched person+clothing pairs.
3. **Only upper body** — Current implementation handles tops/shirts, not full-body garments.
4. **Pose estimation speed** — MediaPipe adds ~1s per image. Could be optimized with caching.